MILITARY SPECIFICATION

POTTED CIRCUIT CARD ASSEMBLY FOR TOW-2/E6 PROBE AND S&A ASSEMBLY

This specification is approved for use within the U.S. Army Armament, Munitions, and Chemical Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification contains the requirements, quality assurance provisions, testing and packaging of an electronic assembly for TOW-2/E6 Probe and S&A Assembly.

2. APPLICABLE DOCUMENTS

2.1 Government documents

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification for

MIL-P-48698 - Probe and S&A Assembly, TOW-2/E6

MIL-R-14293 - Resistor, Fixed, High Megohm (Hermetically Sealed)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army Armament Munitions and Chemical Command, Attn. AMSMC-QA, Picatinny Arsenal, New Jersey 07806-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedure and Tables for Inspection by Attributes
MIL-STD-275	-	Printed Wiring for Electronic Equipment
MIL-STD-750	-	Test Methods of Semiconductor Devices
DOD-STD-2000	-	Soldering Technology, High Quality/ High Reliability

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS

U.S. ARMY ARMAMENT, RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC)

9399722	-	Circuit Card Assembly Potting
9387141	-	TOW 2/E6 Probe and S&A Assembly

(Copies of specifications, standards, drawings, publications and other Government documents required by manufacturers in connection with specific aquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.1.3 Order of precedence. In the event of conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Materials. Materials shall be in accordance with the applicable drawings, specifications, and standards.

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3.2 Components and assemblies. The components and assemblies shall comply with all requirements specified on Dwg. 9399722, all associated drawings, and with all requirements specified in applicable specifications and standards.

3.3 Physical characteristics.

3.3.1 Potting. Potting material shall comply with the requirements of Dwg. 9399723, and shall be manufactured and dispensed in accordance with the processes, procedures, and controls delineated therein. The potted circuit card assembly shall not show evidence of exposed components leads or circuitry but shall expose the five support holes. Bubbles or voids adjacent to a component that do not contact that component for a length greater than the diameter of that component are acceptable. The potted circuit card assembly shall have a shore hardness of 70.

3.3.2 Potted electronic assembly. The potted electronics assembly shall comply with Dwg. 9399722 and as specified in 4.5.4.

3.4 Environmental conditioning. The potted circuit card assembly (dwg. 9399722) shall function properly when exposed to the same environmental conditions (storage temperature, operational temperature, vibration, and shock) as a subassembly to the TOW-2/E6 Probe and S&A Assembly (dwg. 9387141).

3.5 Control drawings. The contractor shall supply the Government a certificate of conformance and an inspection record (per DI-R-1724) for the following specification and source controlled components.

Drawing No.	Part
9388032	Insulation sleeving, electrical
9388034	Insulation sleeving, electrical
9399716	Tubing, cable
9399723	Foam potting compound
9399725	Mold release

3.6 Electronic assemblies performance characteristics.

3.6.1 Timing delay. After tip switch closure, the potted circuit card assembly shall exhibit a 500 ± 50 microsecond delay before a base detonator voltage appears between the white/red cable (E4) and the white/brown cable (E5).

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3.6.2 Tip firing energy. The voltage on the tip S&A detonator shall be 33.3% minimum of the initially applied voltage, 360 microseconds minimum after tip switch closure.

3.6.3 Base firing energy. The voltage on the base S&A detonator shall be 33.3% minimum of the initially applied voltage, 800 microseconds minimum after switch closure.

3.7 Soldering. Soldering procedures and acceptance criteria for the soldered components shall be in accordance with DOD-STD-2000.

3.7.1 Solderability. Electronic assembly parts that are to be soldered shall comply with the solderability requirements of DOD-STD-2000.

3.8 Process control. The contractor shall submit to the Government for approval with the first article sample, process control documents for the processes listed in the following sub-paragraphs. These processes shall be those used in the fabrication of the first article sample without prior approval of the Government. At unscheduled intervals during the operation of the approved processes, the Government inspector shall observe the processes to ascertain that production to the approved process control documents is being maintained. No deviation without approval from the procuring activity shall be allowed. In the event that the contractor desires a change in an approved process control document, a new first article sample may, at the discretion of the Government, be required.

3.8.1 Process control of potting. A process control document shall be prepared for potting the electronic assembly. It shall fully describe the entire process and the controls for maintaining the process. It shall specify the preparation of all materials, pot life, and curing temperature and time.

3.8.2 Process control of soldering. Process control documents/procedures shall be prepared for all soldering operations. These documents shall fully describe the entire process and controls for maintaining the process/procedures in compliance with the soldering requirement of 3.7.

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3.9 Qualified suppliers. A qualified component supplier shall be one who has successfully met the requirements of this specification and first article inspection specified herein.

3.10 First article inspection. This specification contains provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

3.11 Workmanship. All parts and assemblies shall be fabricated and assembled in a thorough, workmanlike manner. They shall be free of burrs, sharp edges, cracks, dirt, grease, rust and other foreign matter. The cleaning method used shall not be injurious to any parts, nor shall the parts be contaminated by the cleaning agents. All required marking and stamping shall be neat and sharply defined.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspections. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this specification.

4.1.1 Responsibility for compliance. All items must meet all requirements of Sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance or defective material.

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4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3)

b. Quality conformance inspection (see 4.4)

4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a first article sample as designated by the Contracting Officer, for evaluation in accordance with the provisions of 4.3.2. The sample shall consist of the items and quantities listed in Table I. In addition, the first article sample shall consist of 36 potted circuit card assemblies to be tested as subassemblies to the TOW-2/E6 Probe and S&A Assembly. First article test samples shall not be considered part of the production quantities.

4.3.2 Inspection to be performed. See MIL-A-48078 and Table I herein. Upon completion of continuity circuit test for first article testing of the TOW-2/E6 Probe and S&A Assembly (MIL-P-48698), remove the potted circuit card assembly from each unit. Inspect the potting material for cracks/deterioration and test the timing delay, tip firing energy, and base firing energy. Only gages or instruments approved by the Government will be permitted to conduct the necessary inspections.

4.3.3 Rejection. See MIL-A-48078.

4.3.4 Completion. Upon completion of first article testing, all components, subassemblies and potted circuit card assemblies shall be sent to: Commander, U.S. Army Armament, Research, Development and Engineering Center, ATTN: SMCAR-AEF-C, Picatinny Arsenal, New Jersey 07806-5000.

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Table I - First article inspection

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-70661 (AR)

PARAGRAPH	TITLE: Circuit Card Assembly, Potting		SHEET 1 OF 1		DRAWING NUMBER: See Below NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Components				
	For each component: examination of defects	10		3.2	Corresponding CD sheet in 4.4.2
	Subassemblies				
	For each subassembly: examination of defects	10		3.2	Corresponding CD sheet in 4.4.2
	Circuit Card Assembly, Potting (Dwg. 9399722)				
	Examination of defects (see note 1)	10		3.2	4.4.2.15
	Potting material (see note 2)	36		3.3	4.5.4
	Timing delay (see note 2)	36		3.6.1	4.5.1
	Tip firing energy (see note 2)	36		3.6.2	4.5.2
	Base firing energy (see note 2)	36		3.6.3	4.5.3

NOTES: 1. Major 107 is not to be performed during first article.
2. Remove the potted circuit card assembly from the TOW-2/E6 Probe and S&A Assembly upon completion of the continuity circuit test (MIL-P-48698) before performing these inspections/tests

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4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. Inspection lots shall comply with formation provisions of MIL-A-48078. Lots shall not exceed three (3) calendar month's production or 5000 units.

4.4.2 Examination. See 4.4.2 of MIL-A-48078. Unless otherwise specified in the Classification of Defects and Test sheets, sampling plans and procedures for major and minor defects shall be in accordance with MIL-STD-105, Inspection level II.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.1	Transistor	1 OF 1	9387177
			NEXT HIGHER ASSEMBLY: 9399674

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL	None defined				
MAJOR					
101	Proper identification or marking		0.40%	3.2	Visual
102	Solderability (see note 1)	10		3.2.1	DOD-STD-2000
103*	Wire bond (see notes 1 and 2)	10		3.2	Pull tester
104*	Internal visual (see notes 1 and 3)	10		3.2	MIL-STD-750 Method 2072
MINOR					
201	Certificate of conformance to dwg.				Certificate
	*To be accomplished by delidding and performing inspections				

NOTES:
Note 1: Use 10 samples per incoming inspection lot.
Note 2: Pull all leads on each device.
Note 3: Internal workmanship.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.2	Resistor, Fixed, Film	1 OF 1	9387179
			NEXT HIGHER ASSEMBLY: 9399674

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL	None defined				
MAJOR					
101	Proper identification or marking		0.40%	3.2	Visual
102	Resistance		0.40%	3.2	MIL-R-14293
103	Solderability (see note)	10		3.7.1	DOD-STD-2000
MINOR					
201	Certificate of conformance to dwg.				Certificate

NOTES:

Note: Use 10 samples per incoming inspection lot.

AMSMC Form 1570, 1 Feb 85

Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-70661 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9387180
4.4.2.3	Semiconductor Device, Diode				NEXT HIGHER ASSEMBLY 9399674
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL	None defined				
MAJOR					
101	Proper identification or marking		0.40%	3.2	Visual
102	Presence of color band on cathode end of diode		0.40%	3.2	Visual
103	Solderability (see note)	10		3.7.1	DOD-STD-2000
MINOR					
201	Certificate of conformance to dwg.				Certificate

NOTES:

Note: Use 10 samples per incoming inspection lot.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-70661 (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.4	Cable, Ribbon	1 OF 1	9388010
			NEXT HIGHER ASSEMBLY 9399695

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL	None defined				
MAJOR					
101	Spark test (see note)		0.40%	3.2	Certificate
102	Film thickness (see note)		0.40%	3.2	Certificate
103	Bond strength (see note)		0.40%	3.2	Certificate
104	Insulation intact (no breaks)		0.40%	3.2	Visual
MINOR					
201	Colored cables arranged in proper sequence		0.65%	3.2	Visual
202	Workmanship		0.65%	3.11	Visual

NOTES:

Note: Certificate of compliance (see 4.3 of dwg. 9388010).

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-70661 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9388028
4.4.2.5	Terminal Lug				NEXT HIGHER ASSEMBLY 9399696
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL	None defined				
MAJOR					
101	Overall height		0.40%	3.2	Gage
102	Width		0.40%	3.2	Gage
103	Outside diameter (2 places)		0.40%	3.2	Gage
104	Hole diameter (2 places)		0.40%	3.2	Gage
MINOR					
201	Protective finish		0.65%	3.2	Visual
202	Workmanship		0.65%	3.11	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-70661 (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.6	Cable, 4 Conductor	1 OF 1	9388031
			NEXT HIGHER ASSEMBLY: 9399696

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL	None defined				
MAJOR					
101	Voltage rating (see note)		0.40%	3.2	Certificate
102	Insulation intact (no breaks)		0.40%	3.2	Visual
MINOR					
201	Presence of proper markings (colors)		0.65%	3.2	Visual
202	Workmanship		0.65%	3.11	Visual

NOTES:

Note: Certificate of compliance (see 4.3 of dwg. 9388031).

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-70661 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9399674
4.4.2.7	Circuit Card Assembly Delay				NEXT HIGHER ASSEMBLY 9399722
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL	None defined				
101	All components properly soldered (see note 3 on dwg.)		100%	3.2	Visual
102	All cables and wires properly soldered (see note 4 on dwg.)		100%	3.2	Visual
103	Proper cable orientation (see note 6 on dwg.)		100%	3.2	Visual
104	Proper wire orientation (see note 7 on dwg.)		100%	3.2	Visual
105	Timing delay		100%	3.6.1	4.5.1
106	Tip firing energy		100%	3.6.2	4.5.2
107	Base firing energy		100%	3.6.3	4.5.3
MINOR					
201	Workmanship		100%	3.11	Visual

NOTES:
Cable - Cable Assy, circuit board (dwg. 9399695)
Wire - Cable Assy, 4 conductor (dwg. 9399696)

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MIL-P-70661 (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.8	Printed Wiring Board Delay	1 OF 1	9399675
			NEXT HIGHER ASSEMBLY: 9399674

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL	None defined				
MAJOR					
101	Proper location of holes (see note 1)	1		3.2	Gage
102	All holes properly plated (see note 6 on dwg.)		0.40%	3.2	Visual
103	Board meets requirements of note 5 on dwg.				Certificate
104	Proper conductive patterns (see note 9 on dwg.)		0.40%	3.2	Visual
MINOR					
201	Diameter of holes (see note 1)	1		3.2	Gage
202	Material (see note 2 on dwg.)			3.1	Certification
203	Workmanship (see note 2)		0.65%	3.11	Visual

NOTES:

Note 1: One sample per incoming lot.
Note 2: Refer to MIL-STD-275

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PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.9	4 - Standoff	1 OF 1	9399676
			NEXT HIGHER ASSEMBLY 9399474

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL	None defined				
MAJOR					
101	Large outside diameter		0.40%	3.2	Gage
102	Outside diameter of knurled section		0.40%	3.2	Gage
103	Inside diameter		0.40%	3.2	Gage
104	Height of base		0.40%	3.2	Gage
MINOR					
201	Threads		0.65%	3.2	Gage
202	Workmanship		0.65%	3.11	Visual

NOTES:

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MIL-P-70661 (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.10	Clamp	1 OF 1	9399678
			NEXT HIGHER ASSEMBLY: 9399674

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL	None defined				
MAJOR					
101	Diameter of holes (2 places)		0.40%	3.2	Gage
102	True position of holes (2 places)		0.40%	3.2	Gage
103	Overall length (Dim "A")		0.40%	3.2	Gage
104	Width		0.40%	3.2	Gage
MINOR					
201	Thickness		0.65%	3.2	Gage
202	Width of center slot (Dim "B")		0.65%	3.2	Gage
203	Workmanship		0.65%	3.11	Visual

NOTES:

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MIL-P-70661 (AR)

PARAGRAPH 4.4.2.11	TITLE Cable Assembly, Circuit Board		SHEET 1 OF 1		DRAWING NUMBER 9399695
					NEXT HIGHER ASSEMBLY 9399674
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL	None defined				
MAJOR					
101	Insulation intact (no breaks)		0.40%	3.2	Visual
102	Continuity of ribbon cable (dwg. 9388010)		0.40%	3.2	Ohmmeter
MINOR					
201	Wire length		0.65%	3.2	Gage
202	Wire separated (see note 5 on dwg.)		0.65%	3.2	Visual
203	Workmanship		0.65%	3.11	Visual

NOTES:

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MIL-P-70661 (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.12	Cable Assembly, 4 Conductor	1 OF 1	9399696
			NEXT HIGHER ASSEMBLY: 9399674

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL	None defined				
MAJOR					
101	Overall length		0.40%	3.2	Gage
102	Length of tubing cable		0.40%	3.2	Gage
103	Length from conductor end to longest insulation sleeve		0.40%	3.2	Gage
104	Continuity of 4 conductor cable (dwg. 9388031)		0.40%	3.2	Ohmmeter
MINOR					
201	Presence of insulation sleeving (3 places)		0.65%	3.2	Visual
202	Proper assembly of parts		0.65%	3.2	Visual
203	Workmanship		0.65%	3.11	Visual

NOTES:

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MIL-P-70661 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.13	Block, Cable				9399712
					NEXT HIGHER ASSEMBLY 9399696
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL	None defined				
MAJOR					
101	Overall length		0.40%	3.2	Gage
102	Overall height		0.40%	3.2	Gage
103	Width		0.40%	3.2	Gage
104	Diameter of mounting hole		0.40%	3.2	Gage
105	True position of mounting hole		0.40%	3.2	Gage
MINOR					
201	Diameter of cable guide hole		0.65%	3.2	Gage
202	Workmanship		0.65%	3.11	Visual

NOTES:

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MIL-P-70661 (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.14	Diode, Zener	1 OF 1	9399719
			NEXT HIGHER ASSEMBLY: 9399674

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL	None defined				
MAJOR					
101	Proper identification of marking		0.40%	3.2	Visual
102	Presence of color band		0.40%	3.2	Visual
103	Solderability (2 leads) (see note)	10		3.2	DOD-STD-2000
MINOR					
201	Certificate of conformance to dwg.				Certificate

NOTES:

NOTE: Use 10 samples per incoming inspection lot.

AMSMC Form 1570, 1 Feb 85

Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-70661 (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.15	Circuit Card Assembly, Potting	1 OF 1	9399722
			NEXT HIGHER ASSEMBLY 9387141

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL	None defined				
MAJOR					
101	Proper encapsulation (see note 5d on dwg.)		100%	3.2	Visual
102	Support holes free of potting material (5 places)		100%	3.2	Visual
103	Potting material		100%	3.3	4.5.4
104	Timing delay		100%	3.6.1	4.5.1
105	Tip firing energy		100%	3.6.2	4.5.2
106	Base firing energy		100%	3.6.3	4.5.3
107	Lot quality confirmation test (see note 1)	36			4.4.3.3
MINOR					
201	Overall height		0.65%	3.2	Gage
202	Height of large step		0.65%	3.2	Gage
203	Height of lower step (support holes)		0.65%	3.2	Gage
204	Contour of assembly (see note 2)		0.65	3.2	Gage
205	Proper color		100%	3.2	Visual
206	Workmanship		100%	3.11	Visual

NOTES:

Note 1: Remove the potted circuit card assembly from the TOW-2/E6 Probe and S&A Assembly upon completion of the continuity test (MIL-P-48698) before performing these inspections/tests.

Note 2: Gage for minimum and maximum contour dimensions.

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4.4.3 Testing. Testing is described in the First Article, Quality Conformance Inspection, and as follows.

4.4.3.1 Quality Conformance. Any item failing to meet the requirements during quality conformance inspection shall be classified defective and removed.

4.4.3.2 Potting material testing. The potting material shall be checked for hardness, compressive strength, flexural strength, dielectric strength, volume resistivity, thermal conductivity, and percent of water absorption in accordance with Dwg. 9399723 on samples taken from each potting material manufacture's lot. One test sample shall be molded for each batch of mixed material which will be used for hardness testing and retained for reference. Identification shall be maintained between the sample and assemblies. Failure of the test specimen to comply with the hardness requirement shall result in rejection of those circuit card assemblies assembled from that batch of potting material.

4.4.3.3 Lot quality confirmation testing. Upon completion of the continuity circuit test for lot quality confirmation testing of the TOW-2/E6 Probe and S&A Assembly (MIL-P-48698), remove the potted circuit card assembly from each unit. Inspect the potting material for cracks/deterioration and test the timing delay, tip firing energy, and base firing energy. If any of these tests fail then the probe lot shall be rejected.

4.4.4 Inspection equipment. The inspection equipment required to perform the examination and tests prescribed herein is described in the "Paragraph Reference/Inspection Method" column in the Classification of Defects and Tests sheets. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-488078 and 6.3 herein.

4.5 Methods of inspection.

4.5.1 Timing delay. Connect a 1.0 K ohm 1% load resistor, simulating the base S&A detonator resistance, between the white/red cable (E4) and the white/brown cable (E5). A power supply (of either 42 $\pm$ 1 volt or 67 $\pm$ 1 volt) through a 47.0 K ohm 10% resistor; and a 0.56 microfarad 10% capacitor charged to the power supply voltage, shall be connected to the (+) black cable (E1) and the (-) red cable (E2) for 400 milliseconds minimum, to allow circuit capacitors to charge. The measurement shall be made after simulating the closure of the tip switch by shorting the black cable (E8) to the brown cable (E12). This test shall then be repeated utilizing the power supply not previously used (42 $\pm$ 1 volt or 67 $\pm$ 1 volt).

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4.5.2 Tip firing energy. Connect a 1.0 K ohm 1% load resisitor, simulating the tip S&A detonator resistance, between the red cable (E9) and the brown cable (E12). A power supply (of either 42 ± 1 volt or 67 ± 1 volt) through a 47.0 K ohm 10% resistor; and a 0.56 microfarad 10% capacitor charged to the power supply voltage, shall be connected to the (+) black cable (E1) and the (-) red cable (E2) for 400 milliseconds minimum, to allow circuit capacitors to charge. The measurement shall be made after simulating the closure of the tip switch by shorting the black cable (E8) to the brown cable (E12). This test shall be repeated utilizing the power supply not previously used (42 ± 1 volt or 67 ± 1 volt).

4.5.3 Base firing energy. Connect a 1.0 K ohm 1% load resistor, simulating the base S&A detonator resistance, between the white/red cable (E4) and the white/brown cable (E5). A power supply (of either 42 ± 1 volt or 67 ± 1 volt) through a 47.0 K ohm 10% resistor; and a 0.56 microfarad 10% capacitor charged to the power supply voltage, shall be connected to the (+) black cable (E1) and the (-) red cable (E2) for 400 milliseconds minimum, to allow circuit capacitors to charge. The measurement shall be made after simulating the closure of a switch (either the tip switch by shorting the black cable (E8) and the brown cable (E12) or the ogive switch by shorting the black cable (E8) to E10). The measurement shall then be made once again by simulating closure of the other switch. This entire procedure shall then be repeated utilizing the power supply not previously used (42 ± 1 volt or 67 ± 1 volt).

4.5.4 Potted electronic assembly. The potted electronic assembly shall not show evidence of voids, bubbles and delaminations exceeding the following criteria:

a. Bubbles within the sprue vent area are acceptable provided they do not contact the top or bottom surface of the printed wiring board.

b. Bubbles exposing any portion of a lead termination or junction are not acceptable.

c. Delaminations of the potting and the printed wiring board are not acceptable.

d. Sprue cutoff shall be flush to 0.04 inches below the surface.

4.5.5 Soldering procedures. All soldering procedures with respect to pre and post cleaning, touchup or rework, and workmanship (as applicable to the requirements in 3.7) shall be submitted to the Government for approval.

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4.5.5.1 Printed wiring board cleaning procedures. Printed wiring board assembly cleaning procedures shall be documented supplied to the Government in accordance with Data Item DI-T-5372 (see 6.4).

4.5.6 Material control. The contractor shall submit to the technical agency listed in 6.4, a material control document. This document shall detail the procedures that will be used to assure that only qualified components, material, and subassemblies from qualified suppliers are used to manufacture the Potted Circuit Card Assembly.

5. PACKAGING - Not applicable; refer to MIL-P-48698.

6. NOTES

6.1 Intended use. The Potted Circuit Card Assemblies covered by this specification are intended for use in TOW-2/E6 Probe and S&A Assembly.

6.2 Ordering data. See MIL-A-48078, paragraph 6.1.

6.3 Submission of designs for approval. (See MIL-A-48078). Submit equipment designs as required to: Commander, ARDEC, ATTN: AMSMC-QAT-I(D), Picatinny Arsenal, New Jersey 07806-5000.

6.4 Submission of process control documents. Process control documents as specified in 3.8 shall be submitted in accordance with DI-P-1604 (Mod) with Attachment 1, to: Commander, ARDEC, ATTN: AMSMC-QAT-B(D), Picatinny Arsenal, New Jersey 07806-5000.

6.5 Data requirements. Deliverable data required by the specification is cited in the following paragraphs.

Paragraph	Data Requirement	Applicable DID
3.5	Quality Inspection Test, Demonstration and Evaluation Report	DI-R-1724
4.5.5.1	Processing Procedure Report	DI-T-5372
6.4	Manufacturing Methods Report	DI-P-1604

The above data items shall be delivered to: Commander, ARDEC, ATTN: AMSMC-QAT-B(D), Picatinny Arsenal, New Jersey 07806-5000.

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6.6 Subject term (key word) listing.

TOW-2/E6 Probe and S&A Assembly
Circuit card assembly potting
Control drawing
Soldering
Process control
Qualified suppliers
Certificate of Conformance
Printed wiring board
Data item description (DID)
Quality conformance
Lot quality confirmation testing

6.8 Contractor. Unless otherwise specified, the word "Contractor", as used throughout this specification, shall mean any supplier or producer of items or material to the Government. This shall include commercial contractors, sub-contractors, Government-Owned, Contractor-Operaterated (GOCO) and Government-Owned, Government-Operated (GOGO) plants.

Custodian:
Army-AR

Preparing activity:
Army-AR

(Project 1336-A541)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-P-70661 (AR)	POTTED CRECUIT CARD ASSEMBLY FOR TOW-2/E6 PROBE AND S&A ASSEMBLY

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

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